FH University of Applied Sciences TECHNIKUM WIEN

PROJECT REPORT

in the Bachelor's programme Sports Engineering and Ergonomics
Course BHF 4 HFSPR – Group Project

Acoustic Localisation of Spikeball Impacts using Piezoelectric Contact Microphones

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Vienna, 22.06.2026

Abstract

Introduction:

With Roundnet (Spikeball™) being a relatively new sport and gaining constant popularity, the competitive landscape is evolving. While most established sports rely on virtual referee assistance utilising sensors and cameras to ensure fair play, such technological monitoring systems are currently absent in Roundnet. This pilot study tries to create a sensor-based algorithm capable of predicting ball impacts on the Spikeball™ net to assist decisions made during the game.

Methods:

To capture the acoustic waves generated by ball impacts, five piezoelectric sensors were attached on the frame of the Spikeball™ TITAN set. By using three different machine learning algorithms, the k-Nearest Neighbours (kNN) regressor, a Multilayer Perceptron (MLP) and a 1-Dimensional Convolutional Neural Network (1D-CNN) the localisation was performed. While the kNN and MLP were trained on 35 extracted features, the 1D-CNN was trained on peak-normalised raw waveforms. All models were trained using cartesian coordinates to facilitate the error calculations.

Results:

A Wilcoxon signed-rank test confirmed a significant difference in localisation accuracy between the evaluated algorithms ($p < 0.001$). The 1D-CNN yielded a mean localisation error of 9.39 cm on the net surface, with a pooled standard deviation (s_p) of 1.30 cm. For direct rim impacts, the mean error measured 30.15 cm (s_p = 4.88 cm). Across the entire dataset, encompassing both net and rim impacts, the global average error of the 1D-CNN resulted in 16.40 cm (s_p =3.01 cm).

Conclusion:

Piezoelectric acoustic based localisation provides a highly consistent and robust proof-of-concept for an assisted referee system in Spikeball™. However, the physical irregularities and non-linear wave dampening of the net currently limit absolute accuracy. To ensure the efficacy of future systems, it is essential that they are trained with larger data sets, providing a wider range of impact angles, strengths, heights and net setups. Future attempts could include a clear differentiation between rim and net hits within the software.

Keywords:

Spikeball; Piezoelectric Sensors; Machine learning; Acoustic Localisation; 1D-CNN; kNN; MLP; Convolutional Neural Network; Impact Detection

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1 Introduction

Roundnet, a sport that has been popularised by Spikeball™ (Kankakee, USA), is a sport in which two teams of two players attempt to impede their opponents' progress towards achieving their next shot by shooting at a net. The rules of the game are analogous to those of volleyball, with each team being permitted three touches between shots at the net (“Spikeball Rules,” 2026). The popularity of Spikeball™ extends beyond its status as a recreational activity, it has also gained traction in educational institutions, particularly in physical education classes, and in competitive tournaments (Threatt et al., 2022, pp. 33–34).

As with every other sport, athletes will try to push the boundaries of the rulebook as far as possible, resulting in discussions about the validity of shots. In Roundnet, the problem lies in the 'pocket area', which is the part of the net closest to the rim and creates rather unpredictable movement of the ball. It is illegal to make a 'pocket shot', where the ball hits close to the rim, when serving the ball, whereas a 'rim shot', where the ball hits the rim directly, is always illegal (“Spikeball Rules,” 2026). It is inevitable that athletes will always strive to win. However, it is not possible for judges to be omnipresent on the field of play. Consequently, other sports have adopted modernised and digitalised systems of monitoring play in addition to the traditional referees. A considerable amount of sporting activities has recently adopted the implementation of video-assisted refereeing (VAR), a technological system that facilitates the re-evaluation of any moment during a match. Plenty of other competitive sports have commenced the integration of a combination of video analysis, various sensors and artificial intelligence to facilitate the acquisition of a live second opinion on shots made during gameplay. Moreover, the extracted data can be utilised for statistical analysis, thereby providing a foundation for enhancing the performance of athletes and teams (Dowsett, 2024; Spitz et al., 2021).



Figure 1: Spikeball™ game using the TITAN set (“TITAN Set,” 2026)

The objective of this study was to develop an assisted referee system for Spikeball™. The proposed methodology involved the utilisation of piezo sensors, which would detect vibrations in the net, thereby enabling the precise calculation of the contact location. As Roundnet is a relatively recent addition to the sporting repertoire, there have been no similar studies conducted to date. However, some studies have been undertaken on metal surfaces or tennis rackets with comparable setups (Liu et al., 2022, pp. 11–12). For instance, one study demonstrated that the impact of a 6 mm diameter impactor on a metal surface, with nine attached piezoelectric elements, resulted in an accuracy range of approximately 8.54 % to 16.69 %, which corresponds to a 32.02–50.25 mm error on a 450 mm x 600 mm x 13.5 mm composite sandwich structure (Sikdar et al., 2022, p. 8).
This study will act as a pilot study trying to detect the 6 cm Ø Spikeball™ - Pro Ball on the TITAN set by using vibration sensors and certain machine learning algorithms.

2 Materials and Methods

The Spikeball™ TITAN set and its frame's flat bottom surface provides the maximal contact area for the application of the piezoelectric elements. It is also the strongest, most durable and modern set in the lineup, making it the obvious choice for a repeatable measurement setup. Spikeball™ sets are composed of five frame pieces, which provide an optimal foundation ("TITAN Set," 2026). Research has demonstrated that a configuration of five piezoelectric elements is consistently more precise than four, particularly at temperatures ranging from 0 to 35 °C (Salmanpour et al., 2017, pp. 10–13).

2.1 Procedure

Each Spikeball™ frame segment was sanded and cleaned, before being equipped with a piezoelectric sensor – specifically the EKULIT EPZ-27MS44("Piezoscheibe - EPZ-27MS44 - EKULIT - Elektrotechnik Karl-Heinz Mauz GmbH," 2021) – aligned to the centre of the bottom surface. To connect these sensors with the national instruments data acquisition system (NI-DAQ, NI USB-6210 National Instruments Corporation - 16 inputs, 16-bit, 250 kS/s), shielded cables were soldered on to the sensors. Initially, an operational amplifier circuit was built to amplify each of the sensors' signals. However, after further testing, this was deemed unnecessary, as the output amplitude provided sufficient resolution for the NI-DAQ. No additional buffer stage was required either, as the NI-DAQ provides an input impedance greater than 10 GΩ. Consequently, the sensors were connected directly to the data acquisition hardware with 10 MΩ resistors in parallel to each channel. This was done to counteract the capacitive properties of piezoelectric sensors, which over time could have resulted in significant voltage drift.
A LabVIEW data acquisition application was created, where the five active channels of the NI-DAQ (Version 2024 Q1 (64 bit) 24.1.1f1, NI, Austin, TX, USA) are continuously read with 50

kHz each. This data is buffered in chunks of 250kS (kiloSamples), after which it is sent to a write block and after the recording is stopped, the voltage data is saved in a .lvm file.

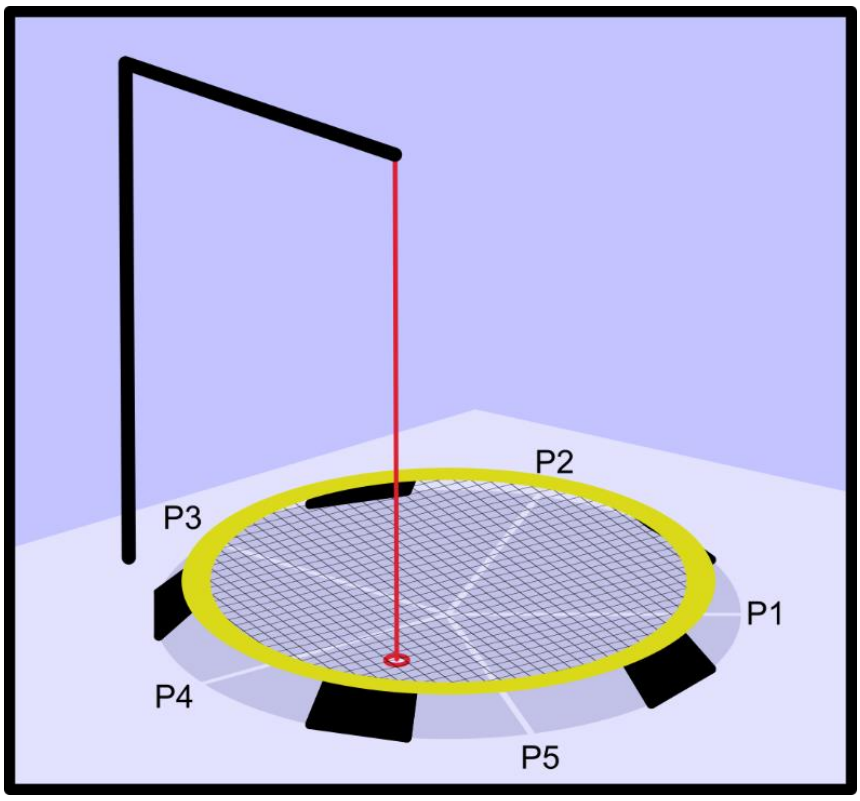


Figure 2: The measurement setup, P1-P5: piezoelectric sensors, red line: a possible position of a plumb line used in the logging of the impact position. The ball drop position is at a predefined height of 1m above the net.

A plumb line attached to a tripod arm was used to transfer locations onto a wooden board, which was placed below the Spikeball™ net. The tripod arm was also used to accurately and repeatedly drop the ball onto the marked position – from a height of 1 metre above the net.
To ensure rigorous model evaluation and prevent data leakage, a total of 30 different impact locations were defined across the net surface, along with ten rim positions for training purposes. The final training dataset comprised 200 measurements, consisting of five recordings at every location. For the statistical analysis, ten impact locations were determined on the net and five on the rim with ten hits recorded each. These 150 independent samples were fed into the trained localisation models. These predicted impact locations, generated by the machine learning models, were then compared against the true impact positions to assess accuracy and reliability of the systems.

2.2 Algorithms and data processing

Three distinct algorithms were implemented to estimate the position of the hits. The first of these was a weighted k-Nearest Neighbour (kNN), the second was a Multilayer Perceptron (MLP), and the third was a one-dimensional Convolutional Neural Network (1D-CNN) which performs automated feature extraction directly from the raw data.

Although the physical setup, the impact locations and our data labels are described using polar coordinates $(r; \theta)$, all machine learning algorithms in this study were trained to predict cartesian coordinates (x, y). This brings up two mathematical problems. First, the polar coordinates have an undefined angle θ when the radius $r = 0$, which would force the model to learn arbitrary angular labels. Second, as standard loss functions such as the Mean Squared Error (MSE) assume a continuous vector space, they are unable to handle cyclical angular values (Bishop, 2006, pp. 105–110). A prediction of $\theta = 359°$ for a target at $\theta = 1°$ represents a physical error of two degrees but results in a disproportionately large penalty in the angular loss calculation, as the MSE is calculated:

$$\boldsymbol{MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \quad \textbf{(1)}$$

Consequently, the MSE is calculated for the specified angles.

$$\boldsymbol{Error_{POL} = \left(\theta_{pred} - \theta_{true}\right)^2 = (359 - 1)^2 = 128164} \quad \textbf{(2)}$$

It has been demonstrated that a distance or radius which never suddenly switches from 360° to 1° does not generate such high errors. Therefore, utilising only distances as coordinates is a more efficient method of calculation, as it does not require the consideration of many cases. In the case of cartesian coordinates, the Pythagorean theorem must be used to calculate the linear distance between the predicted location (x_{pred}, y_{pred}) and the true location (x_{true}, y_{true}).

$$\boldsymbol{Error_{CAR} = \sqrt{(x_{pred} - x_{true})^2 + (y_{pred} - y_{true})^2}} \quad \textbf{(3)}$$

2.2.1 k-Nearest Neighbour and Multilayer Perceptron

To train the kNN and MLP models effectively, a dense feature vector of 35 variables was extracted from the raw sensor data. The features were deliberately selected to map the acoustic and mechanical wave propagation properties of the Spikeball™ net into numerical representations.

The first domain encompasses spatial geometry, captured by 10 Time Difference of Arrival (TDOA) values. These were calculated via a generalised cross-correlation function with phase transform (GCC-PHAT), along with 10 onset time differences. GCC-PHAT is an algorithm used for sound source localisation and TDOA (Lee et al., 2020). These temporal features are critical, as the wave delay between sensors fundamentally correlates with the spatial distance to the impact location. The second domain represents signal attenuation, utilising 5 absolute peak amplitudes and 5 total signal energy values. Due to the high damping properties of the elastic net, closer impacts yield higher energy readings, providing the models with robust spatial hints even when TDOA calculations suffer from noise. Finally, to decouple the geographic impact location from the kinetic impact force, 5 normalised peak amplitude features were included. This normalisation ensures that the algorithms can learn force-invariant spatial ratios, dramatically increasing the generalisation capabilities of the models.

According to the principles of statistical learning, it is critical to avoid the ‘curse of dimensionality’, a phenomenon where high-dimensional spaces become sparse, reducing performance of distance-based algorithms like kNN (Hastie et al., 2009, pp. 22–24). To mitigate overfitting and ensure model stability, established statistical methods, such as the 'Rule of 10', suggest maintaining a minimum of ten independent training samples per predictor variable (Harrell, 2015, pp. 72–74). Although the ratio of 200 available impact measurements for 35 features does not meet this proportion, it remains statistically viable. Some features are physically coupled and thus highly correlated with each other (e.g. peak amplitude and total energy), meaning that the 35 features are not truly independent, justifying the current sample size while retaining the physical interpretability of the features.

The baseline algorithm employed was a k-Nearest-Neighbour (kNN) regressor, implemented utilising the *scikit-learn* library (Pedregosa et al., 2012) in Python 2026.1 (Python Software Foundation, 2026). General data manipulation, array operations, and efficient model serialisation were handled via the standard Python data science stack, including *NumPy* (Harris et al., 2020), *Pandas* (The pandas development team, 2020), and *Joblib* (The joblib developers, 2025).

Prior to model training with the data in place, they were pre-processed. Every single one of the 35 extracted features was normalised by a Z-Score normalisation (Standard Scaler) as the difference between the scales of features was substantial. This type of data extraction and treatment was done before training and before estimating the positions (Hastie et al., 2009, pp. 48–50) as this preprocessing step ensures that all 35 features contribute equally to the distance calculations, preventing features with numerically larger variances from dominating the distance metric.

To estimate the continuous two-dimensional cartesian coordinates (x, y) of the impact, the algorithm evaluates the k closest reference points in the training dataset, with the neighbourhood size empirically set to $k = 3$. Instead of a simple uniform average, a distance-weighted approach was utilised (Dudani, 1976).

The kNN regressor is predicated on instance-based spatial proximity in space, whereas the MLP functions as a more complex model, learning hierarchical, weighted representations of waveforms and dynamics. This facilitates the determination of more complex and non-linear relations between the extracted features. This non-linearity is essential to accurately allow the algorithm to predict positions even though the net is not homogeneous and does not transport waves equally fast in every direction. The MLP is a foundational stepping stone for the 1D-CNN algorithm (Goodfellow et al., 2016, p. 326). By first extracting the same 35 normalised features as in the kNN model, it needs less computational power, as the 1D-CNN works with a complete cutout from more than 1000 samples per channel.
As previously mentioned, being limited to 200 data trials for the training of a neural network with 35 features, the risk of overfitting is immense. The MLP will learn fluctuations of features (e.g. TDOA is off by 0.002 ms -> fluctuation/noise) rather than the actual wave form and details. To counteract this, a dropout regularisation which randomly blocks out a specified percentage of neurons during training was implemented (Srivastava et al., 2014). By randomly deactivating neurons in the network, the model will not be overly reliant on certain features but rather try to source information from all 35 of them to improve generalisation.
The algorithm was optimised via back-propagation using the Adaptive Moment Estimation (Adam) algorithm, which mainly relies on momentum to overcome empty spaces and Root Mean Square Propagation (RMSProp) to adapt learning speed of individual parameters (Kingma and Ba, 2017). In combination with the coordinate transform and the MSE calculations, the algorithm is forced to reduce the distance between predicated and true location continuously.

2.2.2 1D-Convolutional Neural Network

To further improve the accuracy of the results, a 1D-Convolutional Neural Network was implemented. This method, unlike the previously used algorithms, does not require the manual extraction of features. Instead, it is directly built upon the data, which is cut to an empirically optimised window size of 2000 samples, centred around the global peak of the signal to ensure the network captures the relevant context of the acoustic wave rather than memorising local noise. Preventing noise memorisation is a key factor for preventing overfitting. This is done as the CNN can detect those features itself and can handle higher data volume better than the others (Goodfellow et al., 2016, pp. 329–333). In addition to the libraries used for the kNN and MLP (2.2.1), the *PyTorch* framework was used for its strength in machine learning (Paszke et al., 2019). Prior to training, the wave was normalised along the y-axis (amplitude) as this decouples the kinetic force from the data, ensuring the network does not learn absolute impact strength. Similarly to the MLP, the CNN uses standard dropout regularisation from *PyTorch* to discourage overfitting. During training, a specified percentage of neurons will be deactivated for each forward pass, forcing the algorithm to learn patterns (Srivastava et al., 2014).

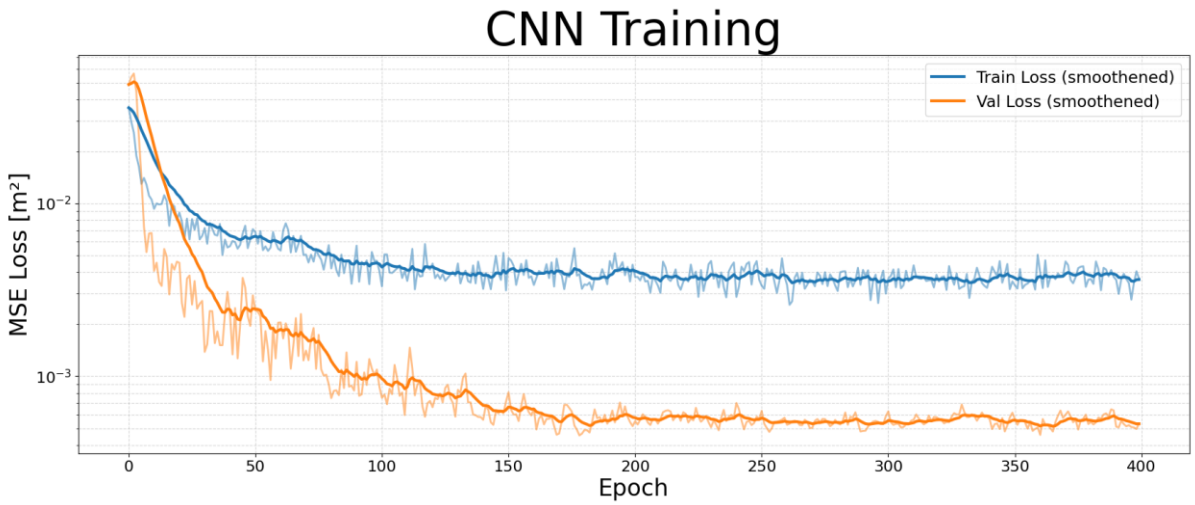


Figure 3: Mean squared error training and validation loss in m² for the CNN training over 400 epochs. The semi-transparent lines represent the raw loss per epoch, while the solid, bold lines indicate the smoothed trends for both the training (blue) and validation (orange) datasets.

Additionally, the algorithm continuously tests for accuracy and progress through auto-validation, as depicted in Figure 3, which demonstrates the MSE over 400 epochs with 10 iterations each. To ensure stable convergence and prevent weight oscillation around the local minimum during the later stages of training, a Step Learning Rate Scheduler (StepLR) was integrated into the pipeline, decaying the learning rate by a factor of 0.5 every 200 epochs. As the network is converging after roughly 200 epochs, the final trained set was stopped at 250 epochs as further training is unnecessary and brings in the risk of overfitting, where the model starts to memorise noise (Goodfellow et al., 2016, pp. 290–292).

3 Results

This chapter presents the empirical findings of the localisation experiments on the Spikeball™ net. The results are divided into two phases. First, a performance comparison of the three machine learning architectures (kNN, MLP and 1D-CNN) is used to evaluate their general accuracy, followed by a statistical validation to determine which algorithm is the most optimal. The second phase is an in-depth analysis of the winning model. Visualising geographic error distributions on the net and isolating systemic bias from the prediction algorithms.

All visualisations presented in this chapter were generated utilising the Python matplotlib (Hunter, 2007) in combination with the SciencePlots package (Garrett and Peng, 2021). Furthermore, all quantitative metrics were extracted using the SciPy library (Virtanen et al., 2020) and the data science stack mentioned in 2.2.1.

3.1 Algorithm comparison

To identify the most suitable machine learning architecture for impact localisation, the localisation performance of the kNN, MLP, and 1D-CNN models was compared using the same evaluation dataset. The comparison focuses on both localisation accuracy and prediction variability. Figure 4 summarises the resulting error distributions and cumulative localisation performance of the three approaches.

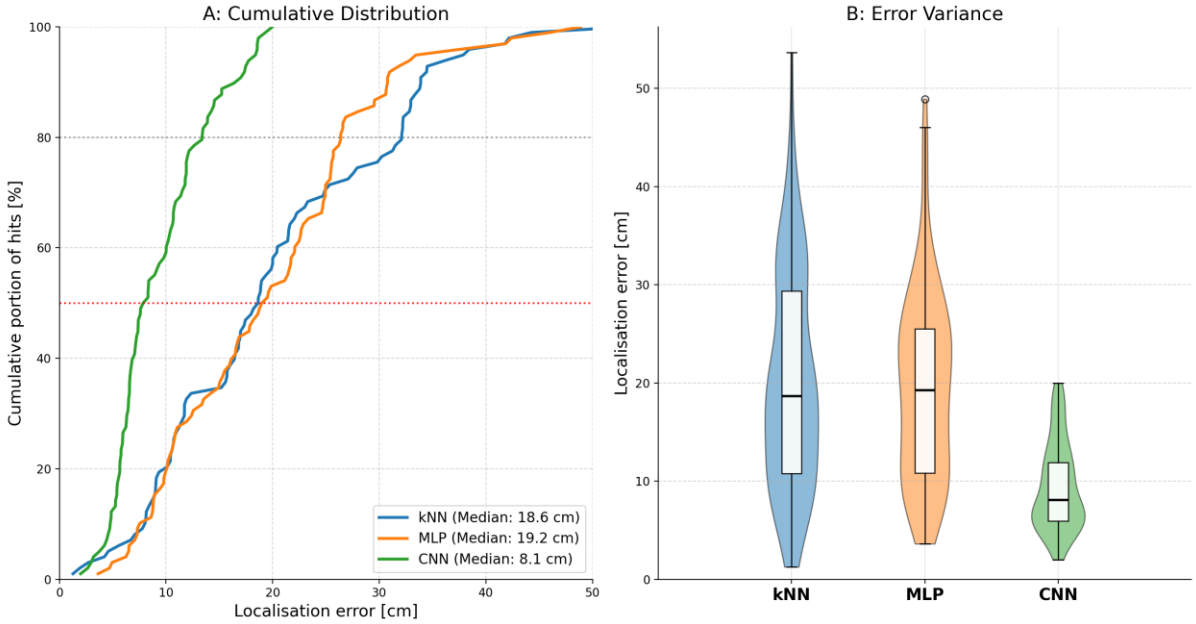


Figure 4: (A) Cumulative Distribution Function (CDF) comparing the localisation error of the kNN (blue), MLP (orange) and 1D-CNN (green) models. The horizontal dashed lines indicate the 50% and 80% cumulative probability levels used for visual comparison. (B) Distribution of localisation errors across the kNN, MLP, and 1D-CNN models. This combined visualisation overlays a standard boxplot, indicating the median (thick black line), interquartile range (white box), and statistical outliers (hollow circles) onto a violin plot, which illustrates a kernel density estimate of the localisation error.

Figure 4A shows the cumulative portion of hits in % for each localisation error in cm for the used machine learning architectures. An initial steep positive slope indicates higher localisation accuracy. The 1D-CNN clearly outperforms the baseline models, achieving a median error of 8.1 cm compared to 19.2 cm for the MLP and 18.6 cm for the kNN. The CNN reaches the 80th percentile threshold at 13.40 cm meaning that 80 % of all predictions have an error less than this margin, whereas the MLP reaches it at 26.39 cm and the kNN at 32.10 cm, using the same dataset. Figure 4B complements this analysis by illustrating the underlying error distributions. The CNN demonstrates a narrower distribution and lower variability in comparison to both baseline approaches, while kNN and MLP exhibit broader distributions with more pronounced upper tails. The interquartile range (IQR) of the CNN is 5.96 cm, compared to 14.71 cm for the

MLP and 18.56 cm for the kNN. Based on the standard $1.5 \times IQR$ threshold for defining statistical whiskers in matplotlib, only the MLP exhibited one outlier with an error of 48.90 cm.

3.2 CNN Results

As demonstrated by both the CDF and the boxplot (Figure 4), the CNN demonstrates the best performance; therefore, this study will further solely analyse the 1D-CNN. Furthermore, a Wilcoxon signed-rank test confirmed that the 1D-CNN significantly outperformed the MLP model ($p < 0.001$), justifying the exclusive focus on the CNN architecture for the subsequent spatial analysis.

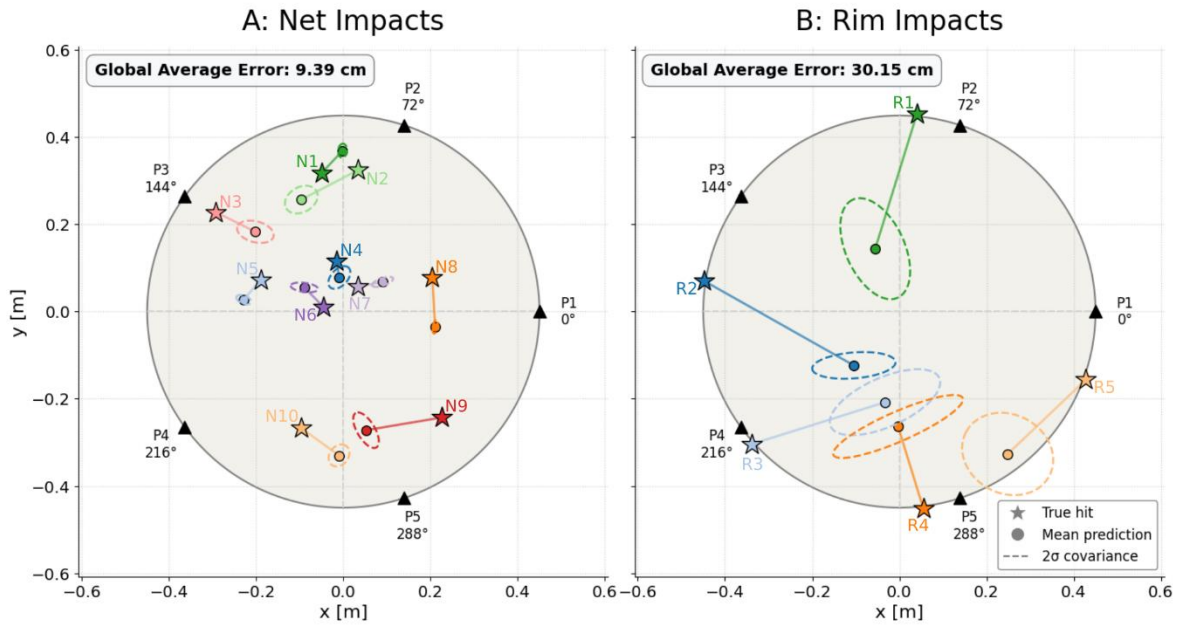


Figure 5: Localisation results of the CNN model for ten predefined net impact locations (left) and five predefined rim impact locations (right), visualised by a top-down view of the Spikeball™ set. Stars denote the true impact positions, while circles indicate the mean predicted locations and triangles the position of piezo. The ±2σ covariance ellipses visualise the spatial distribution of the predictions.

Figure 5 illustrates the localisation performance of the Convolutional Neural Network (CNN) for recorded impact locations on both the Spikeball™ net and rim. The true impact positions are represented by the star markers. Furthermore, ellipses represent the ±2σ covariance regions, visualising the spread of the hit locations predicted by the trained model.

The distance between the true impact position and the mean prediction varied across the tested locations. Across all evaluated measurements, the CNN achieved an average localisation error of 9.39 cm on net hits and 30.15 cm on rim hits. This results in a global average of 16.40 cm.

Table 1: Statistical breakdown of the 1D-CNN localisation error across predefined test locations, separated by impact type net surface: blue, rim: pink, global: purple. N1-N10 and R1-R5 refer to the net and rim hits depicted in Figure 5.

Hit	x_true [m]	y_true [m]	Mean Error [cm]	Std Dev [cm]
N1	-0.05	0.32	7.08	0.71
N2	0.03	0.33	14.80	1.97
N3	-0.29	0.23	10.18	2.16
N4	-0.02	0.12	4.07	1.08
N5	-0.19	0.07	6.00	0.53
N6	-0.05	0.01	6.36	1.16
N7	0.04	0.06	5.76	1.23
N8	0.20	0.08	11.34	0.76
N9	0.24	-0.24	17.79	1.35
N10	-0.10	-0.27	10.86	1.06
Net average			9.39	1.30
R1	0.04	0.45	32.60	5.40
R2	0.06	-0.45	21.03	2.71
R3	-0.34	-0.30	32.12	6.79
R4	-0.45	0.07	39.71	4.09
R5	0.43	-0.16	25.31	4.42
Rim average			30.15	4.88
Global average			16.40	3.01

While Figure 5 visualises the spatial distribution of the CNNs predictions, Table 1 provides the corresponding quantitative breakdown of the errors and their standard deviation for each specific impact location.

To quantify the overall stability of the predictions within each domain, the pooled standard deviation (s_p) was calculated. For all impacts on the elastic net surface, the model achieved a pooled standard deviation of 1.30 cm. In contrast, the pooled standard deviation for the rigid rim impacts deteriorated to 4.88 cm. The mean error ranges from as low as 4.07 up to 17.79 cm for the net shots and for the rim shots the error ranges from 21.03 to 39.71 cm.

4 Discussion

The recorded measurements showed that ball impacts on the Spikeball™ system generate distinguishable vibration signals that can be detected by piezoelectric sensors mounted on the

frame. Analysis of the sensor data revealed that impacts at different locations produced distinct signal characteristics. Throughout the project, several challenges were identified.

4.1 Inhomogeneous wave propagation

The localisation results presented in the previous section revealed systematic deviations between the true and predicted impact positions. To better understand the origin of these errors, the physical wave propagation within the Spikeball™ net must be considered. Due to the non-uniform geometry and tension distribution of the mesh structure, mechanical vibrations are not expected to propagate isotropically. Figure 6 illustrates a simplified example of how wave propagation paths in a ‘perfect’ net may differ from the shortest geometric path.

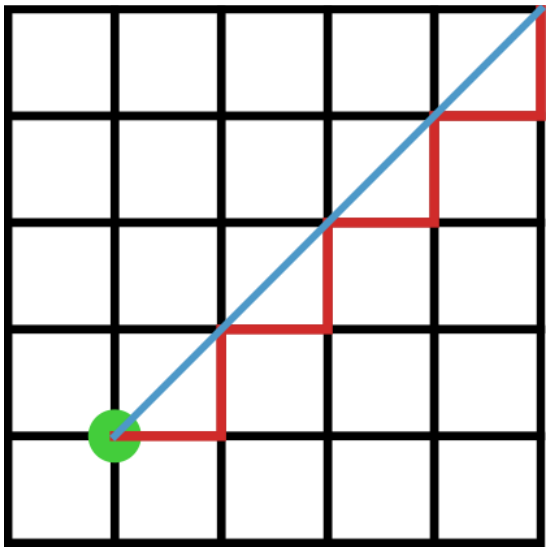


Figure 6: Manhattan Distance - Red Line: possible wave propagation path in a mesh net, Blue Line: Shortest path in homogeneous media

The Spikeball™ TITAN net resembles a rectangular grid with roughly 32 to 38 squares across the diameter of the frame. The geometric structure of the mesh is highly dependent on its radial orientation and the unequal distribution of tension. Due to this warped topology, mechanical waves generated by a ball impact do not propagate uniformly. Instead, the wave follows different effective paths depending on the geometry and tension within the net. For a given impact position, one sensor may receive a vibration through a relatively direct and short mechanical path, while the other may exhibit a complex pathway, similar to the example of a Manhattan distance (Suwanda et al., 2020) depicted in Figure 6.
If the orientation of a net string lines up with two sensors, any ball impact on this connecting line cannot be localised accurately, as this introduces a systematic shift towards the equidistant point between the sensors. This can be explained by higher wave propagation speeds along the axis of tension (Li et al., 2022).

Frame impacts presented unique challenges stemming from material properties, where vibrations were likely dampened through the Spikeball™ set's contact to the ground as well as the higher density and dynamic mass of the frame in comparison to the net. This resulted in data with lower amplitudes and degraded signal-to-noise ratios, potentially explaining the model's inability to accurately detect frame impacts and their locations.

4.2 Temporal resolution

The use of the NI USB-6210 limits the number of samples recorded per channel and second to 50000 ($f_s = 50kHz$). Thus, a temporal resolution of $\Delta t = 2 \cdot 10^{-5}\mathrm{s}$ is achieved.

$$\Delta \boldsymbol{d} = \boldsymbol{v} \times \Delta \boldsymbol{t} = \frac{\boldsymbol{v}}{\boldsymbol{f_s}} \tag{4}$$

As detailed in equation (4), the spatial resolution (Δd) is proportional to the wave propagation speed (v) per sample and is therefore bottlenecked by the NI-DAQs maximum sampling rate. In testing, the TDOA of a ball impact wave between sensors ranged anywhere from 1 to 20 samples, which at these proposed wave speeds and temporal resolution of 50 kS/s/channel leads to a spatial resolution of 0.1 cm at 50 m/s to 4 cm at 2000 m/s. While the mean wave propagation speed was not determined in this project, it nevertheless becomes clear that to achieve more accurate results, a DAQ supporting a higher sampling rate would have to be implemented.

4.3 Training Data Variation

Due to time constraints, a limited number of datasets were acquired for model training purposes. To potentially improve localisation accuracy and future reliability of the measurement setup, further trials could be conducted, while tweaking a set of variables – net tension, net orientation and ball drop height. Based on established deep learning paradigms (Halevy et al., 2009; Sun et al., 2017), it is hypothesized that substantially scaling the training dataset will have a more significant positive impact on spatial accuracy than further increasing the architectural complexity of the algorithm. Additional changes to the measurement setup, such as the automation of ball impact position logging through 2D-video analysis or a custom frame rig, would aid in the practicality of future data acquisition. Alternatively, additional sensors placed around the rim could also aid in increasing the accuracy of predicted results.

4.4 Comparison of Localisation Errors

The results demonstrate that the proposed system based on piezoelectric sensors and a 1D Convolutional Neural Network (CNN) is capable of performing impact localisation on a

Spikeball™ setup. However, the performance is strongly dependent on the region of the impact.

The cumulative distribution function in 3.1 shows that all shots on the net can currently be detected by the convolutional network with an error below 20 cm. However, to reduce this localisation error to a millimetre threshold viable for real-time referee assistance, and to successfully generalise the model to encompass rigid rim impacts, the volume and spatial diversity of the training dataset must be substantially increased.

For impacts on the net, the CNN was able to learn location-dependent features from the vibration signals across ten trials for each position achieving an error of 9.39 ± 1.30 cm. To contextualise this performance, this error can be compared to Sikdar et al.,(2022), who reported a localisation error of 32.02 – 50.25 mm (~8.54 – 16.69 %) error on a 450 mm x 600 mm composite sandwich structure. While direct quantitative comparisons are limited due to differences in normalisation and surface areas, referencing this rigid-body study emphasises the complexity of the Spikeball™ setup. The composite plate utilised by Sikdar et al.,(2022) benefitted from its relative homogeneity, eliminating the complex, anisotropic wave movement mentioned in 4.1. Given this mechanical contrast, achieving a sub-10 cm error (on a structure with a diameter of 91 cm) with a small training set of 200 measurements demonstrates a robust and highly competitive performance by the 1D-CNN.

5 Conclusion

The aim of this project was to investigate whether piezoelectric sensors mounted on a Spikeball™ frame can be used in combination with machine learning methods to estimate ball impact locations based on vibration signals.

Among the evaluated machine learning approaches, the 1D-Convolutional Neural Network (CNN) achieved the best overall localisation performance. For impacts on the net, the model reached a mean localisation error of 9.39 ± 1.3 cm and was able to generate distinct prediction clusters for the tested impact locations. When frame impacts were included, the mean localisation error increased to 16.40 ± 3.01 cm, indicating a reduction in localisation accuracy. Nevertheless, the CNN remained capable of distinguishing between different impact regions and consistently outperformed the MLP and kNN models throughout the study.

Overall, the results confirm that vibration-based impact localisation using piezoelectric sensors and deep learning is feasible. However, the current system does not achieve the accuracy required for precise real-time applications such as refereeing. Instead, it is more suitable for identifying general impact regions rather than exact positions.

Future work should focus on increasing the amount and diversity of training data to enhance localisation accuracy and possibly training the model to specifically differentiate between net and frame impacts based on their specific signal-to-noise profile.

Transparency note

This study was supported by equipment contributions from Kankakee Spikeball Inc. The sponsor was not involved in the design of the study, the collection, analysis or interpretation of data, or the decision to publish. The authors confirm the independence of the content.

Tools of Use:

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List of Abbreviations

Adam	*Adaptive Moment Estimation*
CDF	*Cumulative distribution function*
CNN	*Convolutional Neural Network*
GCC-PHAT	*Generalised Cross-Correlation Function with Phase Transform*
IQR	*Interquartile Range*
kNN	*k-Nearest Neighbours*
MLP	*Multilayer Perceptron*
MSE	*Mean squared error*
NI-DAQ	*National instruments data acquisition system*
RMSProp	*Root Mean Square Propagation*
StepLR	*Step learning rate scheduler*
TDOA	*Time difference of arrival*